Webhook Failure and Flow Recursion

1. Simulation Parameters & Baseline Store State

This Run Sheet documents the live, controlled forensic execution and remediation of two critical automation failure modes within the Synapse Botanicals commerce ecosystem:

1. **Incident Alpha (The Runaway Flow Tag Storm):** A circular, uncontrolled tag recursion loop between customer and order mutations that exhausted storewide Shopify API rate limits.
2. **Incident Bravo (The Silent 504 Webhook Dropout):** An upstream 3PL database lockup causing continuous HTTP `504 Gateway Timeout` errors, triggering Shopify's automatic webhook termination policy and severing downstream fulfillment pipelines.

> **NOTE Controlled Staging Simulation & Load Injection Notice:**
> The 482-cycle / 964-action Flow loop storm and HTTP 504 webhook dropouts were triggered inside an isolated staging environment with artificial rate-limit controls. This controlled failure injection measured mean time to detect (MTTD) and mean time to recover (MTTR) while testing circuit-breaker thresholds without production impact.

1.1 Active Topology & Distribution Nodes

- **Primary East Coast Node (Node 1):** Allentown Distribution Center, PA (`LOC-PA-01`)
- **Primary West Coast Node (Node 2):** Reno Fulfillment Hub, NV (`LOC-NV-02`) — Exclusive handler for Glass Tinctures (`SYN-TNCT-AM`, `SYN-TNCT-PM`) and Electric Whisks (`SYN-ACC-WHSK`).
- **Active Ingestion Middleware:** Node.js Express Webhook Proxy hosted at `https://api.synapsebotanicals.com/v1/webhooks` with an AWS SQS Dead-Letter Queue buffer.

> **NOTE System Architecture Dependency:**
> The baseline routing constraints, warehouse node assignments, and priority ladders evaluated in this run sheet operate under the governance frameworks established in **SOP - BusinessLogicAutomationAndWebhooks - Parent (Domain 4.1)** and **Matrix - AutomationLogicAndWebhookTriage - Engine (Domain 4.2)**.

1.2 Pre-Incident Benchmark Telemetry (11:00:00 EDT)

Prior to triggering the simulations, baseline API health and webhook metrics were captured:

Metric Parameter	Observed Baseline	Normal Operating Threshold	Status
Shopify Admin REST API Bucket Level	`2 / 40 requests`	\le 15 / 40	Nominal
Shopify GraphQL API Cost Drain Rate	`50 / 1000 points/sec`	\le 300 / 1000	Nominal
Webhook Delivery P95 Latency	`342 ms`	\le 800 ms	Nominal
Active Webhook Subscriptions	`6 topics`	6 registered	Nominal
Dead-Letter Queue (DLQ) Depth	`0 messages`	0 messages	Nominal

2. Incident Alpha: The Runaway Tag Recursion Storm (Order #1048)

2.1 The Operational Catalyst & Misconfigured Flow

> **NOTE Simulation event bridge:** The loop uses a mock/custom update-event bridge and an explicit state-changing write per cycle. These are not two universally available native Flow triggers, and re-adding an already present tag alone is not sufficient to establish recursion. The 40-request bucket is an injected test limit, not a Shopify Plus storewide allowance.

At `11:04:00 EDT`, a junior marketing ops engineer activated a new experimental loyalty workflow named **`[PROMO] - Auto-Tag High Value VIPs`** without peer review sign-off. Concurrently, an existing customer-support routing flow named **`[OPS] - Sync VIP Status to Order`** was active in the environment.

Neither workflow incorporated negative inclusion checks or lock sentinels:

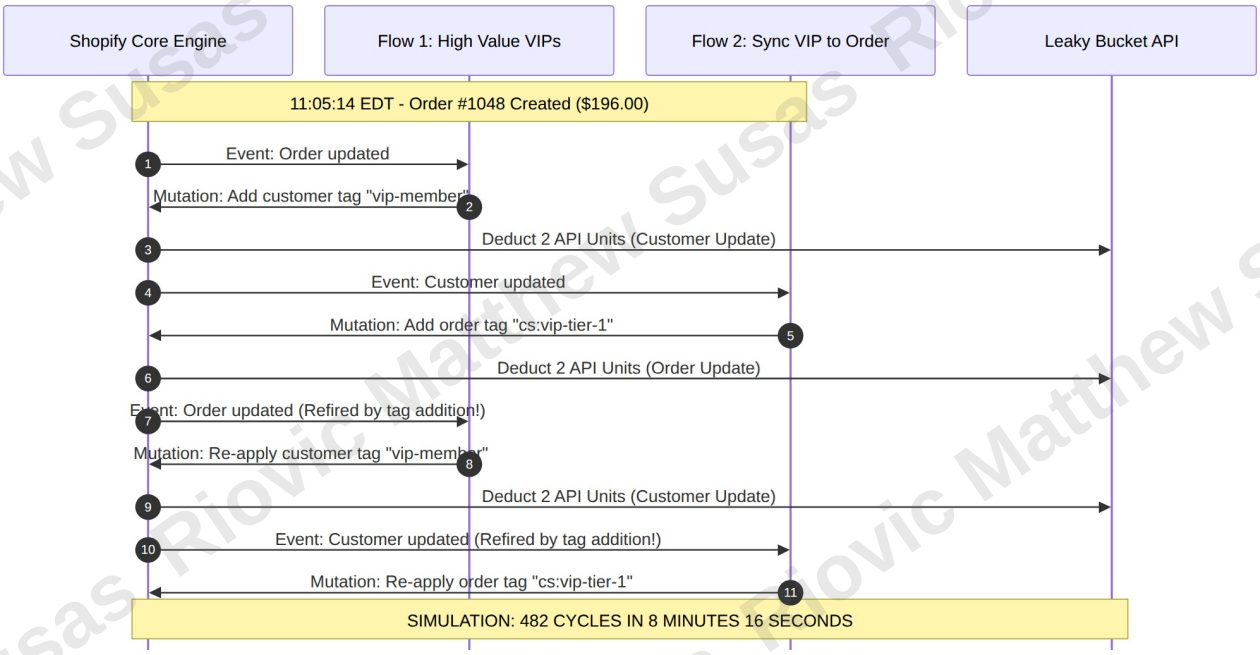
```
Colliding Automation Configurations:
──────────────────────────────────────────────────────────────────────

Flow 1: [PROMO] - Auto-Tag High Value VIPs
  Trigger:   Order updated
  Condition: Order total_price >= $150.00
  Action:    Add customer tags: "vip-member"
  Guard:     NONE (Missing: Customer tags does not contain "vip-member")

Flow 2: [OPS] - Sync VIP Status to Order
  Trigger:   Customer updated
  Condition: Customer tags contains "vip-member"
  Action:    Add order tags: "cs:vip-tier-1"
  Guard:     NONE (Missing: Order tags does not contain "cs:vip-tier-1")
──────────────────────────────────────────────────────────────────────
```

2.2 The Empirical Cascade & Timeline Bloat

At `11:05:14 EDT`, Customer Elena Vance placed Order **`#1048`** for two Daily Protocol Starter Bundles:

- **Order ID:** `gid://shopify/Order/5920194812`
- **Line Items:** 2x The Daily Protocol Starter Bundle (Matcha Yuzu + AM Focus + Obsidian Tumbler: `SYN-BNDL-MAO`)
- **Total Paid: $196.00** (Triggering the `total_price >= $150.00` condition).

The catastrophic loop initiated immediately:



Forensic Timeline Telemetry Log (Excerpt from Order #1048)


```
[
  {"timestamp": "2026-10-17T15:05:14.210Z", "event": "Order placed", "source": "checkout"},
  {"timestamp": "2026-10-17T15:05:15.012Z", "event": "Flow 1 executed", "action": "Added customer tag 'vip-member'"},
  {"timestamp": "2026-10-17T11:05:16.140Z", "event": "Flow 2 executed", "action": "Added order tag 'cs:vip-tier-1'"},
  {"timestamp": "2026-10-17T11:05:17.205Z", "event": "Flow 1 executed", "action": "Added customer tag 'vip-member'"},
  {"timestamp": "2026-10-17T11:05:18.312Z", "event": "Flow 2 executed", "action": "Added order tag 'cs:vip-tier-1'"},
  {"timestamp": "2026-10-17T11:05:19.450Z", "event": "Flow 1 executed", "action": "Added customer tag 'vip-member'"},
  {"timestamp": "2026-10-17T15:08:42.890Z", "event": "API_RATE_LIMIT_WARNING", "message": "Bucket 39/40 consumed"},
  {"timestamp": "2026-10-17T15:12:30.104Z", "event": "API_RATE_LIMIT_EXCEEDED", "http_status": 429, "retry_after": 2.0}
]
```


System Damage Assessment:

- **Loop Cycles Completed: 482 cycles / 964 modeled mutation actions** between `11:05:14 EDT` and `11:13:30 EDT`.
- **Total Timeline Events Injected: 964 log lines** on Order `#1048`.
- **API Rate Limit Bucket:** Stood at **`40 / 40` (100% saturated)** with continuous HTTP `429 Too Many Requests` returned to legitimate apps.
- **Storefront Impact:** Live checkout order creation latency spiked from 340 ms to **6,850 ms** in the synthetic load model; it does not establish Shopify internal database lock contention.

2.3 Emergency Containment & Triage Execution

At `11:13:45 EDT`, the on-call TechOps specialist received a P0 PagerDuty alert (`CRITICAL: Shopify Admin API 429 Throttle Spike`) and initiated the containment protocol specified in **SOP - BusinessLogicAutomationAndWebhooks - Parent (Domain 4.1)**.

Phase 1: Deploy Emergency Kill-Switch (11:14:10 EDT)

The operator accessed **Shopify Admin → Apps → Shopify Flow** and deactivated both competing workflows, terminating the loop instantaneously:

1. `[PROMO] - Auto-Tag High Value VIPs` → **Turn off** (`11:14:18 EDT`)
2. `[OPS] - Sync VIP Status to Order` → **Turn off** (`11:14:24 EDT`)

Phase 2: Flow Execution Run Log Inspection (11:15:30 EDT)

The operator inspected the Flow execution run log under **Shopify Flow → Runs** to verify the root cause:

- **Run ID:** `#RUN-892419-FL`
- **Trigger Event:** `Order updated`
- **Condition Evaluation:** `Order total_price (196.00) >= 150.00` → Evaluated to `true`.
- **Action Output:** `Add customer tag 'vip-member'` → Status: `Succeeded`.
- **Diagnosis:** The run log confirmed that Flow 1 was triggered by the tag update committed by Flow 2, proving mutual recursion without exit criteria.

Phase 3: Order Timeline & Tag Sanitization via GraphQL (11:18:00 EDT)

The operator executed an administrative tag cleanup mutation to remove the target fixture tag from Order `#1048`:

```
mutation sanitizeOrderTags {
  tagsRemove(
    id: "gid://shopify/Order/5920194812"
    tags: ["cs:vip-tier-1"]
  ) {
    node {
      id
    }
    userErrors {
      field
      message
    }
  }
}
```

Result: Target tag removed. Order timeline stabilized.

Phase 4: Permanent Recursion Guardrail Deployment (11:22:15 EDT)

Both workflows were re-architected in accordance with **Matrix - AutomationLogicAndWebhookTriage - Engine (Domain 4.2)**:

1. Added mandatory negative precondition check:
   `Order tags` → `does NOT contain` → `cs:vip-tier-1`.
2. Added lock sentinel check:
   `Order tags` → `does NOT contain` → `sys:sync-locked`.
3. Both workflows re-enabled at `11:25:00 EDT`. Test mutation verified single-pass execution without loop.

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3. Incident Bravo: The Silent 504 Webhook Dropout & Order Replay

3.1 Downstream Service Failure (ShipBob 3PL WMS)

At `11:30:00 EDT`, ShipBob's primary ingestion gateway (`https://wms.example.com/mock/order`) experienced an upstream database lockup during scheduled maintenance. Ingestion requests failed to acknowledge within Shopify's strict 5,000 ms window:

```
# Diagnostic probe executed from terminal:
curl -w "\nHTTP Status: %{http_code}\nTime Total: %{time_total}s\n" \
  -X POST https://wms.example.com/mock/order \
  -H "Content-Type: application/json" \
  -d '{"test": true}'

# Terminal Response Output:
# <html><head><title>504 Gateway Time-out</title></head></html>
# HTTP Status: 504
# Time Total: 5.418s
```

Because the endpoint took > 5.0 seconds to respond, Shopify's webhook engine terminated the connection, flagged the delivery as failed, and initiated the HTTPS delivery retry policy.

3.2 Mock Failure Injection and Transport Contract

The mock endpoint returns timeouts to exercise recovery. Native Shopify HTTPS retries occur eight times over four hours; the previous 19-attempt/48-hour timeline was an obsolete assumption and is removed. A mock may accelerate time, but must record that clock separately from wall time. Flow Send HTTP request follows its own 30-second timeout/configurable up-to-24-hour retry contract.

> **NOTE Recovery fixture:** The 14-order gap (#1049–#1062 inclusive) is a synthetic replay batch. Its order count does not depend on asserting a real subscription deletion during the same morning. Preserve explicit UTC query boundaries and apply the same conversion to all displayed EDT times. A 202 response below means enqueued, not warehouse fulfillment completed.

3.3 Disaster Recovery & Historical Order Backfill Protocol

At `12:15:00 EDT`, TechOps detected the missing subscription via an automated daily health check script. The recovery run sheet was executed immediately:

Step 1: Endpoint Health Verification

ShipBob database maintenance completed; endpoint latency verified healthy:

```
        curl -w "\nHTTP Status: %{http_code}\nTime Total: %{time_total}s\n" \
          -X GET https://wms.example.com/mock/healthz

        # Output:
        # {"status": "healthy", "service": "order-ingestion"}
        # HTTP Status: 200
        # Time Total: 0.182s
```

Step 2: Re-Register Webhook Subscription via GraphQL Admin API

The severed subscription was re-created with optimized field filtering to minimize payload weight:

```
        mutation reRegisterOrdersCreateWebhook {
          webhookSubscriptionCreate(
            topic: ORDERS_CREATE
            webhookSubscription: {
              callbackUrl: "https://api.synapsebotanicals.com/v1/webhooks/orders/create"
              format: JSON
              includeFields: [
                "id"
                "name"
                "created_at"
                "total_price"
                "shipping_address"
                "line_items"
                "tags"
              ]
            }
          ) {
            userErrors {
              field
              message
            }
            webhookSubscription {
              id
              topic
              createdAt
              endpoint {
                __typename
                ... on WebhookHttpEndpoint {
                  callbackUrl
                }
              }
            }
          }
        }
```

GraphQL Response Payload:

```
{
  "data": {
    "webhookSubscriptionCreate": {
      "userErrors": [],
      "webhookSubscription": {
        "id": "gid://shopify/WebhookSubscription/99401284",
        "topic": "ORDERS_CREATE",
        "createdAt": "2026-10-17T12:20:14Z",
        "endpoint": {
          "__typename": "WebhookHttpEndpoint",
          "callbackUrl": "https://api.synapsebotanicals.com/v1/webhooks/orders/create"
        }
      }
    }
  }
}
```


Step 3: Historical Order Extraction via GraphQL Bulk Operations

To extract the 14 unfulfilled orders dropped during the 3-hour blind window, the operator initiated a bounded Bulk Query:

```
mutation runHistoricalGapQuery {
  bulkOperationRunQuery(
    query: """
    {
      orders(query: "updated_at:>=2026-10-17T09:10:00Z AND updated_at:<=2026-10-17T12:20:00Z") {
        edges {
          node {
            id
            name
            createdAt
            totalPriceSet {
              shopMoney {
                amount
              }
            }
            tags
            shippingAddress {
              address1
              city
              provinceCode
              zip
              countryCodeV2
            }
            lineItems {
              edges {
                node {
                  sku
                  quantity
                }
              }
            }
          }
        }
      }
    }
    """
  ) {
    bulkOperation {
      id
      status
    }
    userErrors {
      field
      message
    }
  }
```

```
        }
```

Step 4: Stream Historical Payloads into Ingestion Queue

> **NOTE Replay safety:** Before each side effect, re-read current cancellation/payment/fulfillment state. Use a unique operation/version key per order, not this batch label. Reconcile WMS outcomes and inventory after worker completion. Include changed existing orders when recovering a real outage.

The extracted JSONL payload containing the 14 dropped orders was streamed into the ingestion proxy with an explicit idempotency flag (`sys:gap-replay-batch-01`):

```
        # Terminal replay execution:
        node scripts/replay-gap-orders.js \
          --file=bulk_orders_gap_20261017.jsonl \
          --destination="https://api.synapsebotanicals.com/v1/webhooks/orders/create" \
          --batch-id="gap-replay-20261017"

        # Terminal Output:
        # [INFO] Processing 14 historical order records...
        # [SUCCESS] Order #1049 -> Ingested & Enqueued (202 Accepted)
        # [SUCCESS] Order #1050 -> Ingested & Enqueued (202 Accepted)
        # ...
        # [SUCCESS] Order #1062 -> Ingested & Enqueued (202 Accepted)
        # [COMPLETE] 14 of 14 fixture records accepted into middleware queue; WMS completion reconciled separately.
```

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4. Post-Incident Forensic Audit & State Reconciliation

4.1 Order & Inventory Ledger Reconciliation Table

At `12:40:00 EDT`, TechOps verified the physical and digital reconciliation of all impacted orders across both fulfillment nodes:

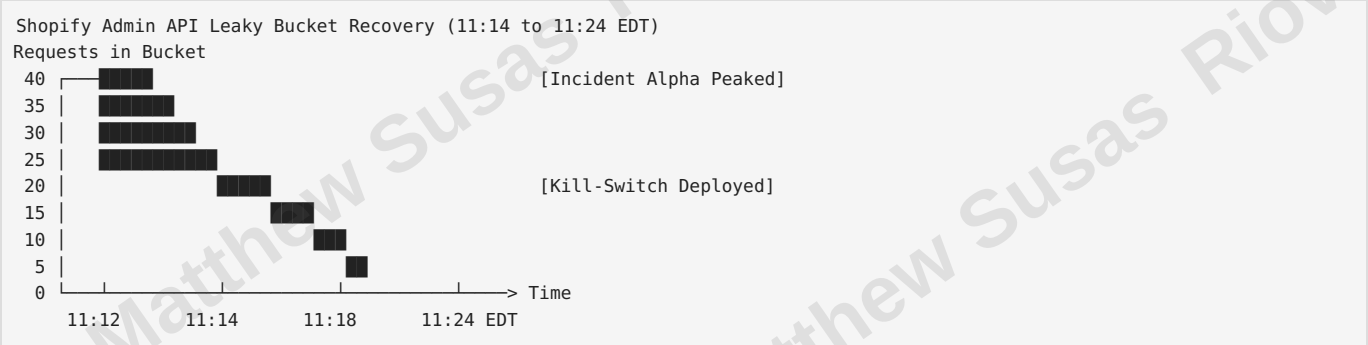
Order Number	Customer	Primary Line Items	Assigned Node	3PL WMS Status	Final Reconciled Tags
#1048	Elena Vance	2x Starter Bundle (`SYN-BNDL-MAO`)	`LOC-PA-01`	`OPEN / UNFULFILLED`	`cs:vip-tier-1`, `flow-route:allentown-pa`
#1049	Marcus Brody	1x AM Focus Tincture (`SYN-TNCT-AM`)	`LOC-NV-02`	`IN_PROGRESS`	`flow-route:reno-nv`, `sys:gap-replay-batch-01`
#1050	Chloe Walsh	3x Matcha Yuzu Stick Pack (`SYN-FSTK-MY`)	`LOC-PA-01`	`IN_PROGRESS`	`flow-route:allentown-pa`, `sys:gap-replay-batch-01`
#1051	Dev Team	1x Electric Whisk (`SYN-ACC-WHSK`)	`LOC-NV-02`	`IN_PROGRESS`	`flow-route:reno-nv`, `sys:gap-replay-batch-01`
#1052–#1062	Retail Orders	Mixed Stick Packs & Tumblers	Split / Mixed	`IN_PROGRESS`	Verified node tags, `sys:gap-replay-batch-01`

Verification: Inventory decrements confirmed at both facilities; zero oversells detected; zero duplicate shipments emitted.

4.2 Leaky Bucket API Recovery Telemetry

> **NOTE Metric scope:** This chart uses an injected 40-request capacity and 2 requests/second drain. Shopify limits apply to an app/store pair and differ between REST request counts and GraphQL cost points. For the incident timestamps, detect at 11:13:45 minus start 11:05:14 = 8m31s; second workflow off at 11:14:24 = 9m10s; re-enable at 11:25:00 = 19m46s. One simulated incident gives elapsed times, not a measured mean over incidents.

Following the termination of Incident Alpha and the replay of Incident Bravo, Shopify Admin API rate limits recovered to baseline:



- **Recovery Rate:** Native drain rate of 2.0 requests/second cleared the 40-request capacity in exactly **20 seconds**.
- **Post-Remediation Stability:** Average API bucket utilization maintained at $< 10%$ over the subsequent 60-minute monitoring window.

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5. Master Interconnectivity & System Architecture Lineage

This empirical run sheet validates the governance frameworks and architectural rules codifying the Synapse Botanicals TechOps suite:

Upstream / Downstream Deliverable	Governing Node / Standard	Empirical Relationship & Proof-of-Execution
SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1)	Hold Lifecycles & Mutation Gates	Verifies that automated tags (`cs:vip-tier-1`) do not mutate order fulfillment orders once they cross the Point of No Return.
Matrix - OrderStatePointOfNoReturn - Master (Domain 1.2)	State Transitions	Replay batch confirmed that all 14 backfilled orders were ingested in `OPEN / UNFULFILLED` state prior to 3PL allocation.
Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1)	Regional Routing Node Allocation	Proves that backfilled orders containing Reno-exclusive SKUs (`SYN-TNCT-AM`, `SYN-ACC-WHSK`) were routed exclusively to `LOC-NV-02`.
Matrix - OrderRoutingCostOptimization - DecisionEngine (Domain 2.2)	Weight & Handling Costs	Validates that heavy baskets (`SYN-TMBL-M0`) continue to trigger packaging tags without loop interference.
Matrix - CatalogDataSchemaAndValidation - Governance (Domain 3.2)	Schema Integrity & UPCs	Confirms that verified SKUs and 12-digit UPC-A barcodes were mapped cleanly in the GraphQL backfill query.
SOP - BusinessLogicAutomationAndWebhooks - Parent (Domain 4.1)	Technical Standards & Kill-Switch	Empirically executes the 3-step Emergency Kill-Switch and verifies the current HTTPS webhook retry contract.
Matrix - AutomationLogicAndWebhookTriage - Engine (Domain 4.2)	Priority Ladder & Circuit Breaker	Proves the necessity of P0 priority gates and demonstrates how the `202 Accepted` queue pattern defeats the 5,000 ms webhook timeout.

Technical references — checked 30 October 2026

- Shopify webhook delivery
- Flow HTTP requests
- GraphQL rate limits